

Phase 14: APIs & Integration — Salesforce

Integration means allowing Salesforce to communicate with another application.

Salesforce Gym System



External App



Payment Gateway / Website / Mobile App

1. REST API

What is REST API?

REST API allows an external application to communicate with Salesforce using HTTP requests.

The main HTTP operations are:

HTTP	Meaning	Salesforce example
GET	Read data	Get customer
POST	Create data	Create customer
PATCH	Update data	Update customer
DELETE	Delete data	Delete customer

Salesforce returns data, usually in JSON:

```
{  
  "Name": "Nikitha",  
  "Phone": "9876543210",  
  "Membership_Status__c": "Active"  
}
```

Simple definition for interview

Salesforce REST API is an API that allows external applications to access and manipulate Salesforce data using HTTP methods such as GET, POST, PATCH, and DELETE

REST API



External system → Salesforce

2. SOAP API

What is SOAP API?

SOAP = Simple Object Access Protocol

It is another Salesforce API used for integration.

SOAP uses only XML format.

External System



SOAP Request



Salesforce



SOAP Response

REST vs SOAP

REST

Lightweight,small

Usually
JSON(preferred),
XML

Easier to use

HTTP-based

Common for
web/mobile apps

SOAP

Heavy weight,Large

XML only

More complex

XML messaging protocol

Common in
enterprise/legacy
integrations

3. Apex REST

Apex REST allows us to create our own REST API using Apex code.

Normally, Salesforce provides standard APIs such as REST API.

But sometimes an external application needs a **custom operation** that isn't directly available in the standard API.

Then you can write an Apex REST class.

External Application



HTTP Request



Apex REST



Apex Class



Salesforce DB



HTTP Response



External Application

For your Gym Management System:

Mobile App



GET /services/apexrest/gym/customer/...



Apex REST Class



Customer Object



Customer details

@RestResource :

This Apex class should be exposed as a REST API .

@RestResource(urlMapping='/gym/customers/*')

global with sharing class GymCustomerAPI {

}

URL mapping means API url path.

@RestResource



"This class is a REST API"

@HttpGet

@HttpGet is used when you want to **retrieve/read data**.

Example:

```
@RestResource(urlMapping='/gym/customer/*')
```

```
global class GymCustomerAPI {
```

```
    @HttpGet
```

```
    global static Customer__c getCustomer() {
```

```
        Customer__c c = [
```

```
            SELECT Id, Name, Phone__c
```

```
            FROM Customer__c
```

```
            LIMIT 1
```

```
        ];
```

```
        return c;
```

```
    }
```

```
}
```

@HttpPost — Create data

Use POST when you want to CREATE a new record.

Example:

```
@RestResource(urlMapping='/gym/customer')
```

```
global class GymCustomerAPI {
```

```
    @HttpPost
```

```
    global static String createCustomer() {
```

```
        Customer__c c = new Customer__c();
```

```
        c.Name = 'Rahul';
```

```
        c.Phone__c = '9876543210';
```

```
        insert c;
```

```
        return 'Customer created successfully';
```

```
    }
```

```
}
```

@HttpPatch — Update data

Use **PATCH** when you want to **UPDATE** an existing record.

Example:

```
@RestResource(urlMapping='/gym/customer/*')
global class GymCustomerAPI {
```

```
    @HttpPatch
    global static String updateCustomer() {

        Customer__c c = [
            SELECT Id, Phone__c
            FROM Customer__c
            LIMIT 1
        ];

        c.Phone__c = '9999999999';

        update c;

        return 'Customer updated successfully';
    }
}
```

@HttpDelete — Delete data

Use **DELETE** when you want to delete a record.

```
@RestResource(urlMapping='/gym/customer/*')
global class GymCustomerAPI {
```



```

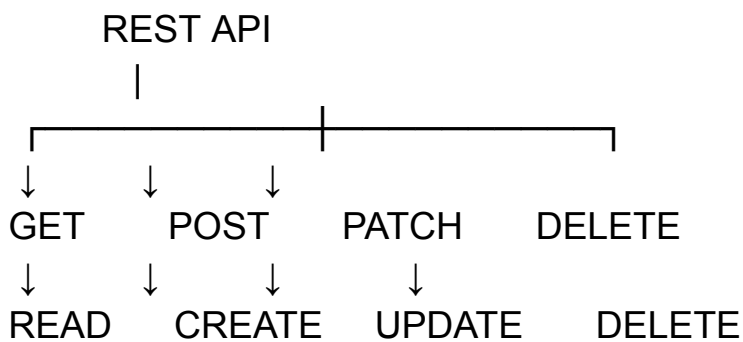
@HttpDelete
global static String deleteCustomer() {

    Customer__c c = [
        SELECT Id
        FROM Customer__c
        LIMIT 1
    ];

    delete c;

    return 'Customer deleted successfully';
}
}

```



One class containing all four :

```

@RestResource(urlMapping='/gym/customer/*')
global class GymCustomerAPI {

    // GET → Read customer
    @HttpGet
    global static Customer__c getCustomer() {

```

```
Customer__c c = [  
    SELECT Id, Name, Phone__c  
    FROM Customer__c  
    LIMIT 1  
];  
  
return c;  
}
```

```
// POST → Create customer  
@HttpPost  
global static String createCustomer() {  
  
    Customer__c c = new Customer__c();  
  
    c.Name = 'Rahul';  
    c.Phone__c = '9876543210';  
  
    insert c;  
  
    return 'Customer created';  
}
```

```
// PATCH → Update customer  
@HttpPatch  
global static String updateCustomer() {  
  
    Customer__c c = [  

```

```
        SELECT Id, Phone__c
        FROM Customer__c
        LIMIT 1
    ];

    c.Phone__c = '9999999999';

    update c;

    return 'Customer updated';
}
```

```
// DELETE → Delete customer
@HttpDelete
global static String deleteCustomer() {

    Customer__c c = [
        SELECT Id
        FROM Customer__c
        LIMIT 1
    ];

    delete c;

    return 'Customer deleted';
}
}
```

12. Standard REST API vs Apex REST

Salesforce REST API

Salesforce has already built the API.

External App



Salesforce REST API



Standard Salesforce Objects

Apex REST

External App



Your Apex REST API



Your Apex logic



Salesforce

Example :

"Give me Customer record 001ABC."

The standard REST API may be enough.

"Get customer + active membership + latest payment + calculate membership status and return everything as one custom response."

A custom **Apex REST** endpoint can make sense.

4. Callouts

What is a Callout?

A callout happens when Salesforce sends a request to an external system.

Example:

Your Salesforce Gym Management System wants to verify a payment using an external payment service.

Salesforce



HTTP Callout



Payment API



Response



Salesforce

In Apex, you can use:

Http

HttpRequest

HttpResponse

Very important distinction

REST API

External system → Salesforce

Callout

Salesforce → External system

A callout is a request made from Salesforce to an external web service or system, usually using HTTP or SOAP.

5. Named Credentials :

You need:

URL

Authentication

Credentials

Security configuration

Why Named Credentials?

Because they help manage:

- **Endpoint URL**
- **Authentication**
- **Security**
- **Credentials**

Apex



Named Credential



Authentication



External API

Named Credential = Secure connection configuration for callouts

6. Connected Apps :

A Connected App is primarily used when an external application wants to connect to Salesforce and access Salesforce resources securely.

Mobile Gym App



Connected App



Salesforce

The Connected App can define things such as:

- **OAuth settings**
- **Scopes**
- **Client credentials**
- **Authentication policies**

7. OAuth Basics

OAuth is extremely important for API integration.

What is OAuth?

OAuth is an authorization framework used to allow applications to access resources securely without sharing the user's password with the application.

Simple example:

User



External App



"Can I access Salesforce?"



Salesforce Authentication



Access Token



External App



Salesforce API

INTEGRATION

